

low-cost temperature sensor and is connected to the analogue input pin of the PIC chip. It can measure temperatures in the range -50°C to 100°C, but in this unit it is mounted on the PCB only to show room temperature.

Circuit and working

Fig. 1 shows the circuit diagram of the alarm clock-cum-temperature indicator. PIC16F73 IC has its reset on pin 1. A resistor and capacitor junction is tied to this pin to perform reset on applying power. The 5V power supply is obtained from a low-voltage transformer (X1), rectifier (BR1) and voltage regulator IC 7805 (IC1).

Pin 20 is V_{cc} and pins 19 and 8 are grounded. Pins 9 and 10 are for connecting a crystal for the oscillator. Any crystal frequency is alright, but it is better to use the highest 20MHz crystal for speed. Two capacitors, each of 15pF, are tied to the end of the crystal to ground. This frequency is called high speed, or HS. There is a configuration register inside the chip, which should be programmed for HS, among other choices, such as XT, RC and LP. The configuration byte we use is 52 hex.

Port B is programmed as output pins. Port A pins are programmed as input pins and pin 2 (AN0) is used as analogue input pin from temperature sensor LM35. Port C pins 4 through 7 are used for driving the anodes of the four LEDs.

Pin C.0 is used as input pin for connecting switch S1 to high logic in order to set the time of clock or alarm time. By continuously pressing it, time changes and the switch has to be released immediately after the required time is displayed.

Pin C.3 is used as another input pin. It is used for setting the alarm time. S2, which grounds this pin, will set the alarm time to be the time currently on display. Thereafter, S1 is to be pressed to adjust to the current clock time.

Pins C.2 and C.1 are used for seconds blinking (not used here) and for

PARTS LIST

<i>Semiconductors:</i>	
IC1	- 7805, 5V voltage regulator
IC2	- LM35 temperature sensor
IC3	- PIC16F73 MCU
DISP1-DISP4	- 7-segment common-anode LED display
T1-T4	- BC558 pnp transistors
ZD1	- 3.3V zener diode
BR1	- DB107 bridge rectifier
<i>Resistors (all 1/4-watt, ±5% carbon):</i>	
R1, R6, R7	- 10-kilo-ohm
R2-R5, R8	- 1.2-kilo-ohm
<i>Capacitors:</i>	
C1, C2	- 0.1µF ceramic
C3, C4	- 15pF ceramic
C5	- 1000µF, 35V electrolytic
<i>Miscellaneous:</i>	
S1-S2	- Tactile switch
VR1	- 10-kilo-ohm trimpot
XTAL1	- 20MHz crystal oscillator
CON1	- 2-pin connector terminal
PZ1	- Piezo buzzer
BATT.1	- 9V battery
X1	- 230V AC primary to 9V, 500mA secondary transformer

clock alarm output. A piezo buzzer of a musical type is connected to it.

Port B pins are used for sinking the currents from the segments of the four LED displays. Segments of all LEDs are connected together and a multiplexed drive lights one LED after another with a delay of 1ms. Multiplexing is done by port C pins 4 through 7, which go low one after another in sequence. This low-going signal pulls the bases of transistors T1 through T4 in sequence, causing current to flow from the collector pins to anode pins of the LEDs. Emitters of the transistors are all tied to 5V line. Transistors are therefore of pnp type. We can use 2N2907 for bright displays; BC557 can also be used.

While signal from the temperature sensor is given to AN0 pin (2), pin 5 is connected to a supply voltage of 2.55V. This is necessary because ADC of the chip converts to values from 0 to 255. Reference voltage is derived from a zener of 3.3V (which has the least temperature variation among all zener values) and a preset. This pin 5 is programmed to be the reference voltage pin for the ADC. This is done by ADCON1 register bits of the chip. By writing 01 to this register, choice of reference voltage for the ADC to work is from pin 5.

Since LM35 gives a linear 10mV output for each degree of temperature, it will output a value of, say, 250mV for 25°C. This will be read by the ADC as 25 because it is one-tenth of the reference voltage. For a temperature of 100°C, it will give $100 \times 10 = 1000\text{mV}$ or 1V. This will be shown as 100 on the ADC. Value of ADC register ADRES is used to show the temperature on the display.

In order to convert the ADC value to decimal, divide the value by ten to get the high digit; balance or remainder will be low digit. First three LEDs show the 3-digit temperature value and the fourth digit is programmed to show °C. Therefore the fourth LED is turned top-down in the PCB while soldering.

It may be noted from the circuit diagram that order of the segments of the LED displays is not in sequence while wiring port B pins. So port B pins 0 through 7 are not wired in sequence to segments a, b, c, d, e, f, g and DP. Rather, order is changed in order to ensure that PCB tracks are easily routed. Order used is b-a-f-g-c-d-e-DP.

It is to be understood that, according to this connection, look-up table for the segment code for each and every digit displayed is to be prepared. For example, to show number 3, segments a, b, c, d and g have to be made low to sink the current and make these glow. As per the above-mentioned order, this gives the code as 0-0-1-0-0-0-1-1. This byte is 23, which is the look-up table entry for number 3.

Software

Code for the clock as well as the temperature indicator was developed using Oshon PIC simulator _IDE software. It provides high-level BASIC for code development and has all peripherals available for simulation of a program under development on the computer screen.

Complete program is in .ASM format and the program code in hEX is